

A Dilution Counterexample to the Quantitative ε -Strong Q -Measure Question

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Status

This packet gives a strengthened conditional counterexample to Question 2 of Aviles–Martinez–Cervantes–Poveda–Saenz [1]. The earlier version assumed a Q -point ultrafilter. The present version only needs a normalized η -strong Q -measure and then gives counterexamples for every $0 < \varepsilon < \eta/2$. In particular, if a strong Q -measure exists, Question 2 has a negative answer for every $0 < \varepsilon < 1/2$.

The packet also records the exact ZFC reduction: any ε -strong measure whose mass on a bounded-per-block ideal is $< 1/2$ already yields the same failure. The remaining limitation is the same-parameter problem for an arbitrary ε -strong measure.

Source Question

The source paper asks the following question in Section 5.

Since $SX \cap X_\mu$ contains all the constant functions, it follows from the last inequality that $P_\mu^{**}(S^{**}z^{**})$ is not L -orthogonal in $(SX) \cap X_\mu$, and thus z^{**} cannot be an L -orthogonal element in X^{**} . $\square$

Combining all the previous claims Theorem 4.10 follows. $\square$

5. OPEN QUESTIONS

After our investigations, one question still remains elusive:

Question 1. *Is there a model of ZFC with $Q^+(\omega)$ -measures but without Q -points?*

This seems plausible, yet hard to arrange. The basic issue is the tension between the existence of $Q^+(\omega)$ -measures and Q -points. Namely, in all the classical models where there are no Q -points there are no $Q^+(\omega)$ -measures either. We also ask whether a quantitative version version of Theorem 4.3 is possible:

Question 2. *Assume μ is an ε -strong Q -measure and $(x_n)_{n \in \omega}$ is an L -orthogonal sequence in a Banach space X . Does*

$$\|x + \mu\text{-}\lim(x_n)\| \geq \|x\| + \varepsilon \|\mu\text{-}\lim(x_n)\|$$

hold for $x \in X$?

In fact, we do not have examples that separate any of the various generalizations of Q -points that have arisen in our work.

In text: if μ is an ε -strong Q -measure and $(x_n)_{n \in \omega}$ is an L -orthogonal sequence in a Banach space X , must

$$\|x + \mu\text{-}\lim(x_n)\| \geq \|x\| + \varepsilon \|\mu\text{-}\lim(x_n)\|$$

hold for every $x \in X$?

Definitions Used

For $\varepsilon > 0$, a finitely additive positive measure $\mu : \mathcal{P}(\omega) \rightarrow [0, 1]$, vanishing on finite sets, is an ε -strong Q -measure if every partition of ω into finite sets has a selector A with $\mu(A) \geq \varepsilon$ [1, Definition 2.1].

For a partition $\mathcal{P} = (P_m)$ of ω into finite sets, write

$$\mathcal{I}_{\mathcal{P}} = \left\{ A \subseteq \omega : \sup_m |A \cap P_m| < \infty \right\}.$$

The source paper defines a compact Cantor space $K_{\mathcal{P}}$ and coordinate functions $f_n \in C(K_{\mathcal{P}})$, and recalls that (f_n) is L -orthogonal [1, Definition 4.4 and Notation 4.5]. It also proves the following estimate [1, Lemma 4.6]: if ν is a positive finitely additive measure and $\nu(I) \leq \delta \|\nu\|$ for every $I \in \mathcal{I}_{\mathcal{P}}$, then

$$\|\nu\text{-}\lim f_n - \|\nu\| \mathbf{1}\| \leq 2\delta \|\nu\|.$$

For comparison with the source hierarchy, μ is δ -fit if for every finite partition there is $A \in \mathcal{I}_{\mathcal{P}}$ with $\mu(A) \geq \delta \mu(\omega)$, and it is fit if this holds for all $\delta < 1/2$ [1, Definition 4.7]. Thus the first theorem below is exactly the quantitative non-fit mechanism behind the counterexample.

Conditional Dependencies

The ZFC reduction theorem has no extra set-theoretic dependency. The relative counterexample theorem assumes the existence of a normalized η -strong Q -measure. This assumption is weaker than requiring a Q -point when $\eta = 1$: a Q -point gives a strong Q -measure, but the source paper also shows under $\mathfrak{d} = \text{cov}(\mathcal{M})$ that atomless strong Q -measures exist [1, Theorem 3.1].

There is a genuine ZFC barrier to proving existence of such measures outright. The source proves that it is consistent with ZFC that there are no $Q^+(\omega)$ -measures [1, Theorem 1.2]; more explicitly, Corollary 3.15 states that in Laver's model, the Mathias model, and Miller's model there are no measure $Q^+(\omega)$ -ideals. Since every ε -strong Q -measure is automatically a $Q^+(\omega)$ -measure, one cannot get a ZFC construction of an ε -strong measure that works in all models.

Proof Intuition

The source estimate says that in the canonical $C(K_{\mathcal{P}})$ test space, a measure whose bounded-per-block sets have mass $< 1/2$ has a μ -limit close to the constant function 1. Then $x = -1$ makes the left side small, while the proposed quantitative lower bound is still larger than 1.

The dilution trick manufactures exactly this situation from any sufficiently strong source measure. Put a small coefficient $t < 1/2$ on an η -strong measure σ , and put the remaining mass on a block-density measure λ that vanishes on every $\mathcal{I}_{\mathcal{P}}$ -set. The $t\sigma$ part preserves enough selector mass to be ε -strong whenever $\varepsilon < t\eta$, but the bounded-per-block ideal sees only at most t mass.

Theorem 1 (ZFC non-fit reduction). *Let $0 < \varepsilon$, and let $\mu : \mathcal{P}(\omega) \rightarrow [0, 1]$ be a positive finitely additive measure vanishing on finite sets. Suppose that μ is an ε -strong Q -measure and that there are a partition $\mathcal{P} = (P_m)$ of ω into finite sets and a scalar $\delta < 1/2$ such that*

$$\mu(I) \leq \delta \mu(\omega) \quad (I \in \mathcal{I}_{\mathcal{P}}).$$

Then Question 2 fails for μ : there are a Banach space X , an L -orthogonal sequence $(x_n) \subseteq X$, and $x \in X$ such that

$$\|x + \mu\text{-}\lim x_n\| < \|x\| + \varepsilon \|\mu\text{-}\lim x_n\|.$$

Proof. Set $M = \mu(\omega)$. Since μ is ε -strong, $M \geq \varepsilon > 0$. Let $X = C(K_{\mathcal{P}})$, and let $x_n = f_n$ be the coordinate functions from the source construction. The sequence (f_n) is L -orthogonal. By the source estimate,

$$\|\mu\text{-}\lim f_n - M\mathbf{1}\| \leq 2\delta M.$$

Write $u = \mu\text{-}\lim f_n$. Since $2\delta < 1$,

$$\|u\| \geq M - \|u - M\mathbf{1}\| \geq M(1 - 2\delta) > 0.$$

Take $x = -M\mathbf{1} \in X$. Then

$$\|x + u\| = \|u - M\mathbf{1}\| \leq 2\delta M < M,$$

whereas

$$\|x\| + \varepsilon \|u\| \geq M + \varepsilon M(1 - 2\delta) > M.$$

Therefore the proposed inequality is false. □

Theorem 2 (Dilution counterexample). *Assume there is a normalized η -strong Q -measure $\sigma : \mathcal{P}(\omega) \rightarrow [0, 1]$, so $\sigma(\omega) = 1$. For every $0 < \varepsilon < \eta/2$, there is an ε -strong Q -measure μ for which Question 2 fails.*

Proof. Choose t with $\varepsilon/\eta < t < 1/2$. Let $\mathcal{P} = (P_m)_{m \in \omega}$ be a partition of ω into nonempty finite sets with $|P_m| \rightarrow \infty$. Let $\mathcal{V}$ be any free ultrafilter on ω , and define

$$\lambda(A) = \lim_{m \rightarrow \mathcal{V}} \frac{|A \cap P_m|}{|P_m|} \quad (A \subseteq \omega).$$

Then λ is a positive finitely additive probability measure vanishing on finite sets. Also $\lambda(I) = 0$ for each $I \in \mathcal{I}_{\mathcal{P}}$, because $|I \cap P_m|$ is bounded while $|P_m| \rightarrow \infty$.

Define

$$\mu = t\sigma + (1 - t)\lambda.$$

This is again a positive finitely additive probability measure vanishing on finite sets.

We first verify that μ is ε -strong. Let $\mathcal{R}$ be any partition of ω into finite sets. Since σ is η -strong, there is a selector S for $\mathcal{R}$ with $\sigma(S) \geq \eta$. Hence

$$\mu(S) \geq t\sigma(S) \geq t\eta > \varepsilon.$$

Thus μ is ε -strong.

If $I \in \mathcal{I}_{\mathcal{P}}$, then $\lambda(I) = 0$, so

$$\mu(I) = t\sigma(I) \leq t.$$

Choose δ with $t < \delta < 1/2$. The preceding theorem applies and gives the desired failure of Question 2. □

Corollary 1. *If a strong Q -measure exists, then Question 2 has a negative answer for every $0 < \varepsilon < 1/2$. In particular, the previous Q -point counterexample is the special case where the strong Q -measure is the ultrafilter measure of a Q -point.*

Proof. Normalize the strong Q -measure. It is then 1-strong, so the dilution counterexample theorem applies with $\eta = 1$. $\square$

Remark 1 (What this does not close). *Starting from a particular ε -strong measure, the dilution argument produces counterexamples only for smaller coefficients, namely any $\varepsilon' < \varepsilon/2$ after normalization. A same-parameter counterexample for every given ε -strong measure would require a new argument showing that the measure itself, or a same-strength modification of it, has the non-fit/block-smallness property above. This is closely tied to the source paper’s Question 3, which asks whether the notions $Q^+(\omega)$ -measure, δ -fit, fit, ε -strong, and strong Q -measure are equivalent or distinct.*

Verification Notes

The proof has two scalar requirements. In the reduction theorem, $\delta < 1/2$ gives

$$2\delta M < M < M + \varepsilon M(1 - 2\delta).$$

In the dilution theorem, $t < 1/2$ gives the block-smallness constant, while $\varepsilon < t\eta$ preserves ε -strongness. The included script

`code/check_scalar_inequality.py`

checks representative scalar choices; the mathematical proof above does not depend on computation.

Limitations and Novelty Check

This remains a conditional packet because it does not prove, in ZFC, the existence of an η -strong Q -measure. Such a proof is impossible in the naive existence sense, since the source paper gives models with no $Q^+(\omega)$ -measures, and ε -strong implies $Q^+(\omega)$. The result also does not address coefficients $\varepsilon \geq 1/2$, where the present block-smallness contradiction cannot work.

I searched the run indexes for arXiv:2501.06537, L -orthogonal sequences and elements, Q -measures, ε -strong Q -measures, fit Q -measures, and the quantitative Theorem 4.3 question; only this packet hit. A bounded web search on 2026-06-16 for exact phrases including “epsilon-strong Q -measure”, “fit Q -measure”, “2501.06537”, and “quantitative version Theorem 4.3 L -orthogonal” did not reveal a later answer to Question 2.

Human Review Recommendation

Review the dilution step $\mu = t\sigma + (1-t)\lambda$, especially the two inequalities $\mu(S) \geq t\eta > \varepsilon$ for arbitrary finite partitions and $\mu(I) \leq t < \delta < 1/2$ for $I \in \mathcal{I}_{\mathcal{P}}$. The remaining mathematical frontier is the same-parameter case for an arbitrary ε -strong measure, not a gap in the proved relative result.

References

- [1] A. Aviles, G. Martinez-Cervantes, A. Poveda, and L. Saenz, *A Banach space with L -orthogonal sequences but without L -orthogonal elements*, arXiv:2501.06537, 2025.